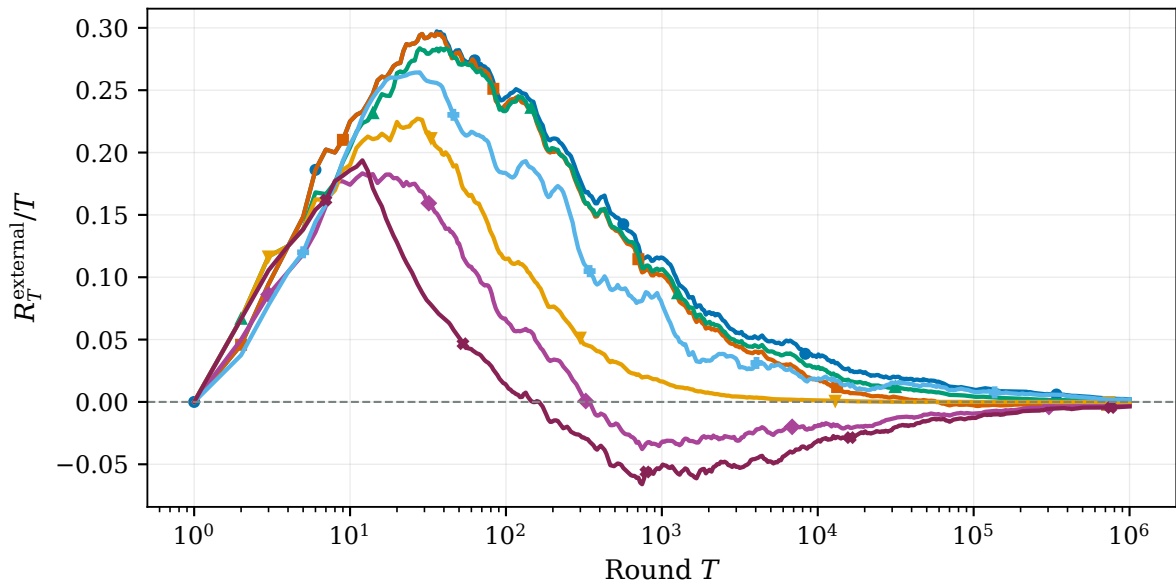


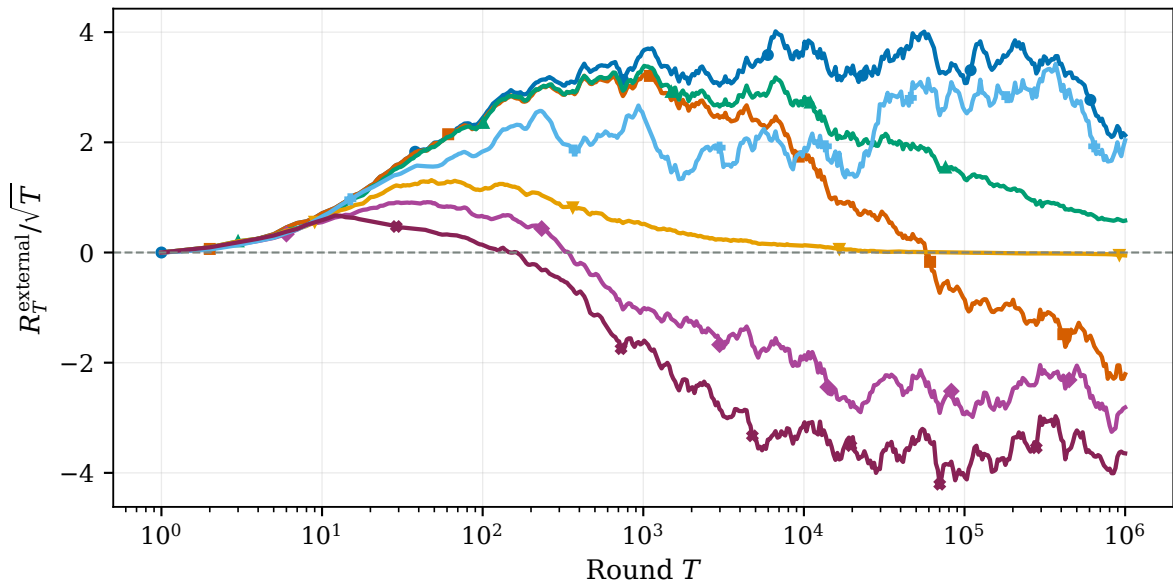
Experiment Information

Figure: Algorithm comparison
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.00212853, SD = 0.000752986, SE = 0.000217368
BM-OptHedge: mean = -0.00220859, SD = 0.00143739, SE = 0.00041494
Hedge: mean = 0.000578758, SD = 0.0001043, SE = 3.01089e-05
Ito-Hedge: mean = -0.00281221, SD = 0.00150603, SE = 0.000434753
OptHedge: mean = -5.24417e-05, SD = 4.36894e-05, SE = 1.2612e-05
RM: mean = 0.00203804, SD = 0.00230171, SE = 0.000664446
SRM: mean = -0.00364579, SD = 0.00159322, SE = 0.000459922



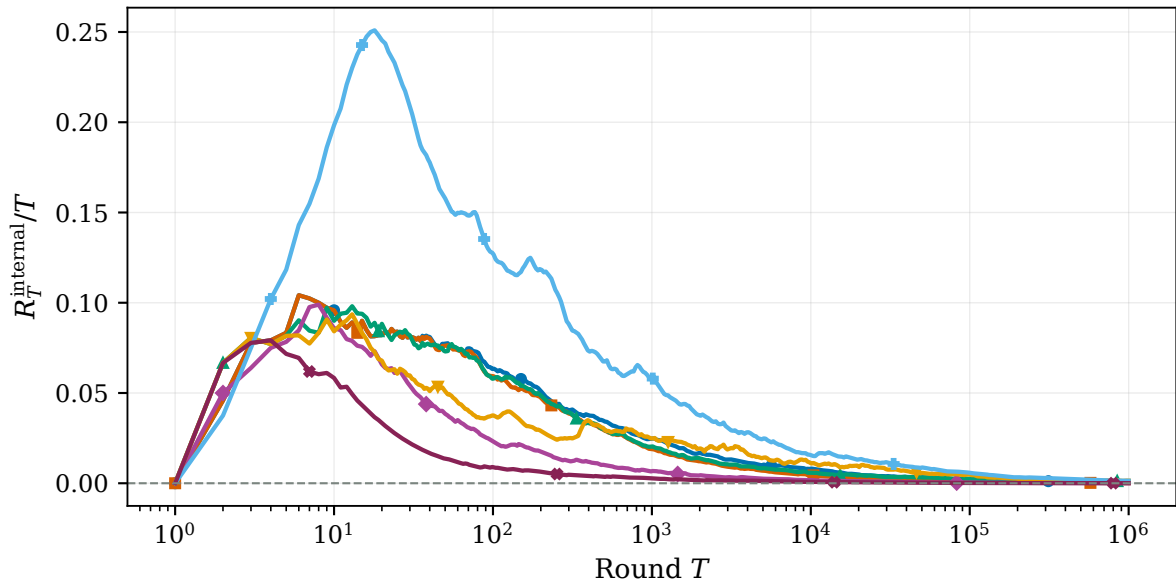
Experiment Information

Figure:	Algorithm comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 2.12853, SD = 0.752986, SE = 0.217368
BM-OptHedge:	mean = -2.20859, SD = 1.43739, SE = 0.41494
Hedge:	mean = 0.578758, SD = 0.1043, SE = 0.0301089
Ito-Hedge:	mean = -2.81221, SD = 1.50603, SE = 0.434753
OptHedge:	mean = -0.0524417, SD = 0.0436894, SE = 0.012612
RM:	mean = 2.03804, SD = 2.30171, SE = 0.664446
SRM:	mean = -3.64579, SD = 1.59322, SE = 0.459922



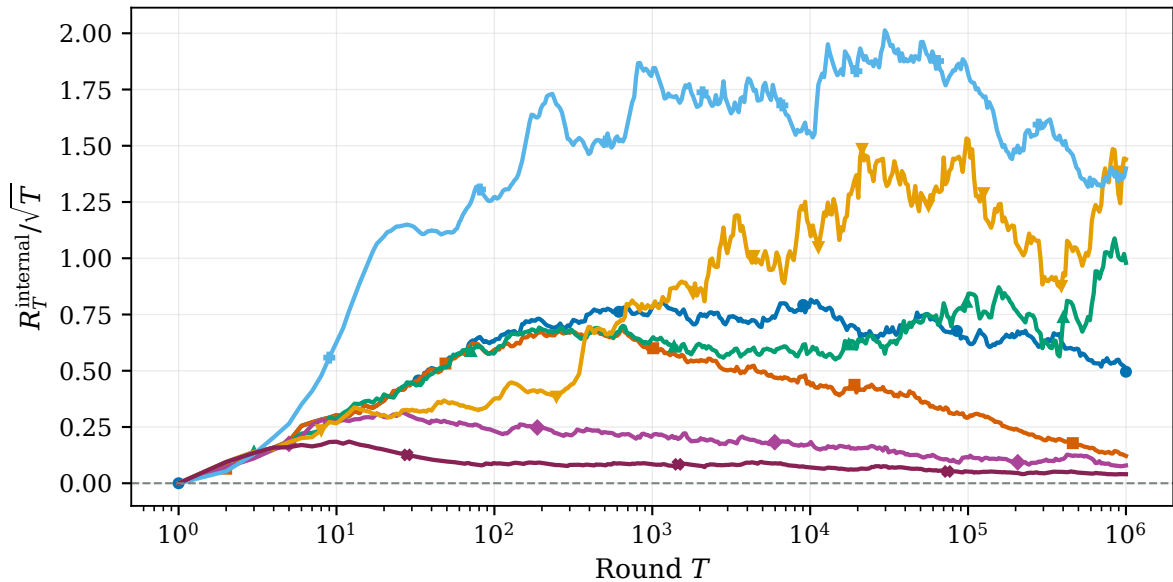
Experiment Information

Figure: Algorithm comparison
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.000495717, SD = 8.48667e-05, SE = 2.44989e-05
BM-OptHedge: mean = 0.000121275, SD = 4.25657e-05, SE = 1.22877e-05
Hedge: mean = 0.000979592, SD = 0.000657893, SE = 0.000189917
Ito-Hedge: mean = 7.895e-05, SD = 4.12671e-05, SE = 1.19128e-05
OptHedge: mean = 0.00144024, SD = 0.000797076, SE = 0.000230096
RM: mean = 0.0013993, SD = 0.000615364, SE = 0.00017764
SRM: mean = 4.0175e-05, SD = 1.14256e-05, SE = 3.29828e-06



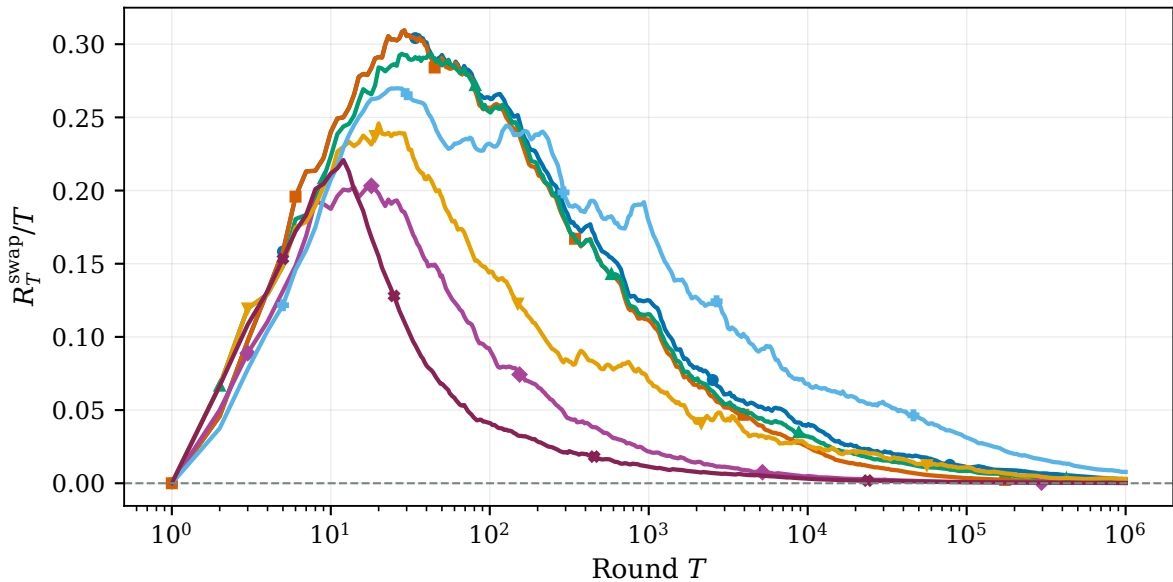
Experiment Information

Figure:	Algorithm comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.495717, SD = 0.0848667, SE = 0.0244989
BM-OptHedge:	mean = 0.121275, SD = 0.0425657, SE = 0.0122877
Hedge:	mean = 0.979592, SD = 0.657893, SE = 0.189917
Ito-Hedge:	mean = 0.07895, SD = 0.0412671, SE = 0.0119128
OptHedge:	mean = 1.44024, SD = 0.797076, SE = 0.230096
RM:	mean = 1.3993, SD = 0.615364, SE = 0.17764
SRM:	mean = 0.040175, SD = 0.0114256, SE = 0.00329828



Experiment Information

Figure: Algorithm comparison
Environment: Lazy random walk
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.00267633, SD = 0.000439674, SE = 0.000126923
BM-OptHedge: mean = 0.000410117, SD = 6.23952e-05, SE = 1.80119e-05
Hedge: mean = 0.00234382, SD = 0.00112066, SE = 0.000323507
Ito-Hedge: mean = 0.000187867, SD = 9.92245e-05, SE = 2.86437e-05
OptHedge: mean = 0.00298063, SD = 0.00159729, SE = 0.000461098
RM: mean = 0.00776705, SD = 0.00323116, SE = 0.000932755
SRM: mean = 0.000185383, SD = 7.08558e-05, SE = 2.04543e-05



Experiment Information

Figure:	Algorithm comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 2.67633, SD = 0.439674, SE = 0.126923
BM-OptHedge:	mean = 0.410117, SD = 0.0623952, SE = 0.0180119
Hedge:	mean = 2.34382, SD = 1.12066, SE = 0.323507
Ito-Hedge:	mean = 0.187867, SD = 0.0992245, SE = 0.0286437
OptHedge:	mean = 2.98063, SD = 1.59729, SE = 0.461098
RM:	mean = 7.76705, SD = 3.23116, SE = 0.932755
SRM:	mean = 0.185383, SD = 0.0708558, SE = 0.0204543

